

Liming Zheng

Aerial Robotics Researcher | Aerospace Engineering

Bio-inspired aerial robotics • Morphing aircraft • Autonomous perching

✉ l.zheng-1@tudelft.nl ✉ limingzheng01@gmail.com
🏢 Faculty of Aerospace Engineering, TU Delft 🌐 limingzheng.com
🔧 BioMorphic Intelligence Lab • Micro Air Vehicle Lab (MAVLab)



Aerial robotics researcher specializing in bio-inspired and morphing aircraft. My work spans the full development cycle of aerial robots, from morphology-driven design and aerodynamic characterization to flight control and real-world experiments. I am particularly interested in how adaptive morphology and physical intelligence can enable agile, robust, and environmentally interactive flight.

RESEARCH FOCUS

Morphology-driven intelligence Bio-inspired aerial robotics Morphing UAVs Adaptive aerodynamics
Autonomous perching Flight dynamics and control

EDUCATION

- 2022 – now** **Ph.D. in Aerospace Engineering**
Delft University of Technology, the Netherlands
Thesis: *Morphology-Driven Design of Bio-Inspired Aerial Robots for Adaptive Flight and Perching*.
Supervisors: Guido de Croon, Salua Hamaza, and Alexander van Zuijlen
- 2018 – 2021** **M.Sc. in Control and Simulation**
Northwestern Polytechnical University, China
Thesis: *Autonomous Soaring in Turbulent Environments Using Reinforcement Learning*.
- 2014 – 2018** **B.Sc. in Airworthiness**
Northwestern Polytechnical University, China
Thesis: *Optimal Transition Corridor Planning for VTOL Fixed-Wing Aircraft*.

JOURNAL ARTICLES AND PREPRINTS

- 2026** **A squirrel-inspired drone with enhanced stability, agility and maneuverability via whole-body morphing**
Liming Zheng, Alexander van Zuijlen, Salua Hamaza
Nature Communications **17**, 6365. DOI: [10.1038/s41467-026-72822-w](https://doi.org/10.1038/s41467-026-72822-w)
- 2026** **Head-body coupling inspired by gliding mammals restores pitch stability and enables aerodynamic braking for perching**
Liming Zheng, Baihui Chen, Alexander van Zuijlen, Salua Hamaza
Under review at the *Proceedings of the National Academy of Sciences (PNAS)*.
- 2026** **Patagium and tail morphology shape aerodynamic performance and control authority in gliding-mammal-inspired wings**
Liming Zheng, Baihui Chen, Alexander van Zuijlen, Salua Hamaza
Under review at *Bioinspiration & Biomimetics*.

- 2026** **Tactile memory enables active semantic exploration and sensor deployment by tiny drones**
Chaoxiang Ye, **Liming Zheng**, Guido de Croon, Salua Hamaza
Under review at *IEEE Transactions on Robotics*.
- 2024** **Albero: Agile landing on branches for environmental robotics operations**
Liming Zheng, Salua Hamaza
IEEE Robotics and Automation Letters, 9(3), 2845–2852.
- 2020** **A novel control mode of bionic morphing tail based on deep reinforcement learning**
Liming Zheng, Zhou Zhou, Pengbo Sun, Zhilin Zhang, Rui Wang
arXiv preprint. [arXiv:2010.03814](https://arxiv.org/abs/2010.03814)

CONFERENCE PAPERS

- 2026** **Cooking beyond frames: A stereo event camera dataset in the kitchen**
Chengming Feng, Hesam Araghi, **Liming Zheng**, Julien Dupeyroux, Xucong Zhang, Jan van Gemert, Nergis Tomen
European Conference on Computer Vision (ECCV), accepted for oral presentation.
- 2025** **The influence of wing and tail morphology on the aerodynamics of gliding mammals**
Liming Zheng[†], Baihui Chen[†], Alexander van Zuijlen, Salua Hamaza
International Symposium on Adaptive Motion of Animals and Machines (AMAM). [†]Equal contribution.
- 2022** **Using deep reinforcement learning to improve the robustness of UAV lateral-directional control**
Rui Wang, Zhou Zhou, Xiaoping Zhu, **Liming Zheng**
33rd Congress of the International Council of the Aeronautical Sciences (ICAS), 5489–5504.
- 2020** **Application of vector jet control technology on aircraft longitudinal control**
Liming Zheng, Zhiliang Bai, Zhou Zhou, Rui Wang
32nd Congress of the International Council of the Aeronautical Sciences (ICAS).

SELECTED AWARDS AND ACHIEVEMENTS

- 2024** **First Place, International Micro Air Vehicle Competition (IMAV)**
- 2024** **XPRIZE Rainforest Bonus Prize**
- 2019** **Top Prize, Challenge Cup academic competition**
- 2019** **First Prize, 5th China Post-Graduate Future Flight Vehicle Innovation Competition**
- 2018** **Second Prize, 15th China Post-Graduate Mathematical Contest in Modeling**
- 2018** **First Prize, Dassault AERO-Design Innovation Civil Aircraft Design Challenge**
- 2017** **6th Place, Air Cargo Challenge (ACC), Zagreb**
- 2015** **National Third Prize, China Aeromodelling Design Challenge**

RESEARCH AND TECHNICAL EXPERTISE

Aerial robotics and control

Bio-inspired robot design; autonomous perching; flight dynamics; trajectory planning; feedback control; flight testing

Aerodynamics and experiments

CFD; wind-tunnel testing; aerodynamic characterization; fixed-wing and high-angle-of-attack flight

Programming

MATLAB; Python; C++; C; ROS; ROS 2

Design and prototyping

CATIA; SolidWorks; AutoCAD; Fusion 360; lightweight aircraft design; hands-on platform integration

Flight experience

More than ten years of drone piloting, aircraft building, and experimental flight operations

Languages

Mandarin Chinese; English